

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and 10+ years developing high-performance numerical algorithms and simulation software for relativistic plasmas. Author of the open-source Paramo library for kinetic particle evolution and radiative transfer. Experience with MHD codes on leadership-class HPC systems and with machine learning for calibrating physical models. Peer-reviewed publications, competitive grants, and collaboration across interdisciplinary teams.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, Lagrangian particle dynamics, numerical PDEs (advection-diffusion, Fokker-Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Geospatial & Remote Sensing: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

Languages: Python, C/C++, Fortran 95, R, SQL, Shell, Rust, Java, Julia

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Education

Ph.D. in Physics

Universitat de València, Spain

ADVISOR: Prof. Miguel A. Aloy & Dr. Petar Mimica.

THESIS: *Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows*

Oct. 2011 – Jul. 2017

Excellent *cum laude*.

M.Sc. in Physics

Universidad Michoacana de San Nicolás de Hidalgo, Mexico

ADVISOR: Prof. José A. Cervera

THESIS: *Study of TOV stars with the SPH method*

Aug. 2009 – Sep. 2011

B.Sc. in Physics

Universidad Autónoma del Estado de México, Mexico

ADVISOR: Prof. Francisco S. Guzmán

THESIS: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Aug. 2004 – Dec. 2008

“Dr. Juan Josafat Pichardo” Award

Experience

Independent Contractor

Mercor, Seattle, WA, USA

Jan 2026 — Present

- Physics expert for a leading AI lab: reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.

Spatial Data Scientist

TealWaters, Seattle, WA, USA

May 2025 — Jan 2026

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 imagery and terrain derivatives for ML land-cover and wetland models; built Python tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer

Jan 2024 — Present

Self-directed, Seattle, WA, USA

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker–Planck equation coupled to radiative transfer; published in ApJ (2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.

Software Engineer

Apr 2022 – Jan 2024

Paychex, Rochester NY, USA; based in Seattle, WA, USA

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts for production incidents; prepared and validated Oracle EBS data with SQL.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.

Postdoctoral Research Associate

Feb 2021 – Apr 2022

Rochester Institute of Technology, Rochester, NY, USA

- Led a 7-person team upgrading **PatchworkMHD**, a C code for magnetohydrodynamic simulations of supermassive black hole binaries, in an NSF-funded project.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Designed and ran benchmark simulations of the new features on TACC’s *Frontera* supercomputer (about 10,000 node-hours), identifying performance optimization opportunities.
- Calibrated Paramo model parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron-star mergers [6, 7] and a third on blazar heating and cooling [8].

Postdoctoral Research Fellow

Oct 2018 – Nov 2020

Purdue University, West Lafayette, IN, USA

- Architected **Paramo**, an open-source Fortran 95 library that solves the Fokker–Planck equation coupled to radiative transfer; parallelized and optimized it with OpenMP.
- Paramo has been adopted by other groups, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) and supported the finding that blazar jets are launched with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Modeled the COVID-19 outbreak in Mexico (Python) and co-wrote Spanish-language infographics against misinformation.

Postdoctoral Research Fellow

Jan – Sep 2018

Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacan, Mexico

- Built a Python pipeline that turned GRTrans black-hole simulation images into training data for an SVM classifier of black-hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for radiative-transfer spectra and light curves.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant

Oct 2011 – Jul 2017

Universitat de València, Burjassot, Valencia, Spain

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the **C-SPEV** simulation code.
- Automated runs of the **C-SPEV** radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Analyzed Fermi-LAT and VLBA multi-wavelength observations, cross-matching sources across catalogs and fitting log-parabola spectra.
- Fit **C-SPEV** models to telescope data, finding spectral and light-curve patterns that quantify plasma magnetization in blazars.
- Extended **C-SPEV** to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

Publications

Articles

[12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.

[11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., *Random Forest Link Landscape Position to Drought*

Resilience in Giant Sequoia Forests, in prep.

- [10] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, **ApJ** **976**, 182, (2024), [arXiv:2405.17581](#).
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, **MNRAS** **513**, 5766–5779, (2022), [arXiv:2201.07790](#).
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, **Phys. Rev. D** **106**, 083015, (2022), [arXiv:2112.09817](#)
- [6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, **ApJ** **919**, 95, (2021), [arXiv:2106.05356](#)
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, **MNRAS** **501**, 4092–4102, (2021), [arXiv:2009.02273](#).
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, **MNRAS** **496**, 974–986, (2020), [arXiv:1910.14049](#).
- [3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, **MNRAS** **468**, 1169–1182, (2017), [arXiv:1612.06383](#).
- [2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, **MNRAS** **438**, 1856–1869 (2014), [arXiv:1310.5441](#).
- [1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, **Phys. Rev. D** **80**, 084023 (2009), [arXiv:1009.1250](#).

Proceedings

5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, **Astron. Nachr.**, 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), [arXiv:2011.13797](#).
4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, **Astron. Nachr.**, 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), [arXiv:2011.13805](#).
3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, **PoS(SWIFT 10)** **233**, 159 (2014), [arXiv:1502.07882](#).
2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, **EPJ Web Conf.** **61**, 02007 (2013), [arXiv:1309.4612](#).
1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, **J. Phys.: Conf. Ser** **454**, 012001 (2013), [arXiv:1211.1794](#).

Research Grants

NASA Fermi Cycle-12 Guest Investigator Program

A simple model to understand the blazar sequence, PI: Giannios, D., Co-I: **Rueda-Becerril, J. M.**

Grant #121077

2019

Grants and Fellowships

Oct. 2018 – Nov. 2020: Fellowship from the Mexican Federal Government for international postdoctoral studies awarded by the National Council of Science and Technology (CONACyT).

Jan. – Sep. 2018: Fellowship from the Mexican Federal Government under the *Program for the Professional Development of Higher Education Institutions*, awarded by the Secretariat of Public Education.

Sep. 2014 – Aug. 2016: Fellowship from the Mexican Federal Government to study abroad awarded by the National Council of Science and Technology (CONACyT).

Oct. 2011 – Jun. 2014: Fellowship *Santiago Grisolia* awarded by the Council of Education, Research, Culture and Sport of the Valencian Community, Spain.

Sep. 2009 – Aug. 2011: Fellowship for MSc studies at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo, granted by the Mexican Council of Science and Technology (CONACyT).

Jun. – Aug. 2007: Fellowship for a temporary stay (3 months) in a national research center under the *XVII summer of scientific investigation program* awarded by the Mexican Academia of Science.

Projects

Blazar Population ML

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

github.com/altjerue/ilwikak-ml

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Developing forecasting pipeline for blazar gamma-ray flares using a time series foundation model, fine-tuned with synthetic light curves generated with Tleco.

WindsOfChange

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++) github.com/UW-ForestMycobiomeLab/WindSporesHills

- Developing C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrains with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library (Python/Rust)

github.com/zkdavis/Tleco

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONACyT Funded)

github.com/altjerue/paramo

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.
- Prototyped GPU acceleration using OpenACC directives on core Fokker-Planck solver kernels.

PatchworkMHD

Feb 2021 — Apr 2022

Large-Scale MHD Simulation Code (NSF Project)

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

Awards

Marcos Moshinsky Award: for *Best Poster* presented at the IWARA 2020 Video Conference, Mexico City, 6 – 12 September 2020.

Lic. Juan Josafat Pichardo Cruz Award: for finishing and defending a licentiate thesis within a year after completing the undergraduate credits, granted by the Universidad Autónoma del Estado de México, 2009.

Invited Talks

The Role of Machine Learning in Environmental Science

Seminar on Climate Change Ecology, University of Washington, Seattle, WA, USA, March, 2024

Morphology of the spectra from numerical simulations of the internal shocks model for blazars

Astrophysics Seminar, Purdue University, West Lafayette, IN, USA, February 4, 2019

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars

DATA group weakly Seminar, Instituto de Astronomía, UNAM, Mexico City, Mexico, June 19, 2018

Numerical treatment of non-thermal radiation in the internal shocks model for blazars

Weekly Seminar, Instituto de Física y Matemáticas, Morelia, Mexico, March 2, 2018

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars

IVICFA's Fridays: Computation in Physics, IFIC, Paterna, Spain, October 17, 2014

Meetings and conferences

Contributed Talks.....

Simulations of supermassive binary black holes accretion dynamics in the spinning case

APS April Meeting 2022, New York City, NY, April 9–12, 2022

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows

APS April Meeting 2021, April 17–20, 2021

The blazar sequence revised

9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020

<https://www.youtube.com/watch?v=BAZNWLNT69M>

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario

Extreme-Astrophysics in an Ever-Changing Universe, Ierápetra, Greece, June 16–20, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets

XXXIV Biennial meeting of the Royal Spanish Society of Physics, Valencia, Spain, July 15–19, 2013

Poster Sessions.....

A numerical approach for radiative cooling in relativistic outflows

9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020

Marcos Moshinsky Award for Best Poster. <https://www.youtube.com/watch?v=0TJiKg7kOPI>

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars

Swift: 10 years of Discovery, Rome, Italy, December 2–5, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets

The Innermost Regions of Relativistic Jets and Their Magnetic Fields, Granada, Spain, June 10–14, 2013

Teaching & Mentoring Experience

Zachary Davis [8, 9]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2022

Amanda O. Harrison [5]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2020

Hao Zhang [4]

Graduate student, Department of Physics and Astronomy, Purdue University

Mentoring

2018 – 2019

Thermodynamics (Graduate Level)

Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo

Substitute Instructor

Jun 2018

Professional Development

High Performance Computing on Frontera

Jason Allison et al., TACC, Austin, TX, USA, May 20, 27 and June 3, 2021

Lecture

Writing Winning Grants

Dr. Lauren Broyles, Purdue University, West Lafayette, IN, USA, November 7, 2019

Lecture

XSEDE HPC Workshop: Summer Boot Camp

John Urbanic, Purdue University, West Lafayette, IN, USA, June 3 – 6, 2019

Workshop

Data Analysis and Machine Learning with Python

Dr. Alejandro Torres, Universitat de València, Burjassot, Spain, February 7 – 16, 2017

Workshop

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit

Dr. Vassilios Mewes, Universitat de València, Burjassot, Spain, July 6 – 7, 2016

Workshop

No. of hours: 8

The Universe in the Light of PLANCK and BICEP2

Prof. Nick Mavromatos, Universitat de València, Burjassot, Spain, May 23 – 16, 2014

Lecture series

No. of credits: 2

Dark Matter

Prof. Alejandro Ibarra, Universitat de València, Burjassot, Spain, September 23 – 27, 2013

Lecture series

No. of credits: 2

International Cargèse School on Cosmic Accelerators

Institut d'Études Scientifiques de Cargèse, Cargèse, France, April 23 – May 8, 2013

Summer school

Introduction to C++ Programming

Dr. Jacek Generowicz, Universitat de València, Burjassot, Spain, April 9 – 12, 2012

Workshop

No. of credits: 6

Numerical Relativistic Astrophysics

Prof. Luciano Rezzolla, Universitat de València, Burjassot, Spain, March 27 – April 4, 2012

Lecture series

No. of hours: 9

Fortran for Scientific Computing

HLRS, University of Stuttgart, Stuttgart, Germany, Mar. 5 – 9, 2012

Workshop

No. of hours: 33

Outreach

Los más rápidos y los más furiosos (*The Fastest and the Most Furious*)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco
Tabasco, Mexico

Online talk

September 4, 2020

Una simulación de la física y la astrofísica (*A Simulation of Physics and Astrophysics*)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco
Tabasco, Mexico

Online talk

August 14, 2020

ANITA y la teoría de los universos paralelos (*ANITA and the Theory of Parallel Universes*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/anita

Blog post

May 29, 2020

¿Qué es el modelo SIR? (*What is the SIR Model?*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/modelo-sir

Blog post

May 25, 2020

Evolución del brote epidémico de COVID-19 (*Evolution of the COVID-19 Epidemic Outbreak?*)

Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post/covid19

Blog post

April 5, 2020

Collaborator with the data analysis/modeling

Annual Department of Physics and Astronomy Poster Event

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Posters (3) presentation

November 13, 2019

Post-Doc Panel Q&A: What Happens When we Complete our PhDs?

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Panelist

April 10, 2019

Annual Department of Physics and Astronomy Poster Event

Department of Physics and Astronomy, Purdue University
West Lafayette, IN, USA

Poster presentation

November 14, 2018

¿Decía Einstein la verdad? (*Was Einstein Telling the Truth?*)

Facultad de Ciencias, Universidad Autónoma del Estado de México
Toluca, Mexico

Talk

March 11, 2009

Synergetic Activities

X Scientific Meeting of the Spanish Astronomical Society

Organizing contributor, 14–16 December, 2012

Valencia, Spain

Other activities

Aug 2007– May 2009: Representative of the Physics students community at the Governing Council of the Faculty of Sciences of the Universidad Autónoma del Estado de México.

Languages

Spanish: Native proficiency

English: Full professional proficiency

Catalan: Basic

Nahuatl: Basic